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Selective modification of skin barrier lipids

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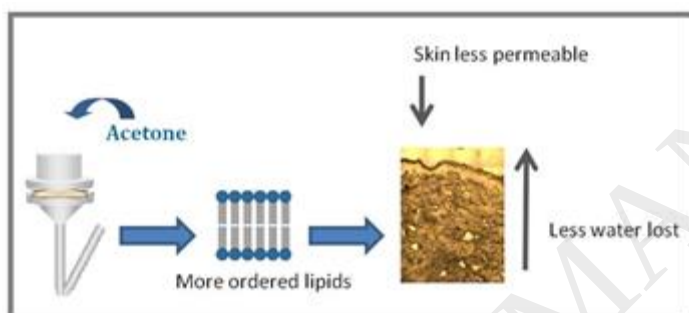
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Graphical abstract



Highlights

- More ordered SC lipid organization after Ac treatment
- Better understanding of the role of the different skin lipids
- Acetone treatment decreases skin TEWL
- Select lipid solubilization can modulate SC permeabilization.

Abstract

The molecular organization of stratum corneum (SC) lipids is important for maintaining the barrier properties of the skin. The majority of intercellular lipids are in a solid state at normal humidity (RH) and ambient temperature; however, several studies indicate that a small fraction exist in a fluid state. In a previous work, a preferential solubilization of fluid skin lipids by acetone (Ac) was envisaged. A different change in the skin permeability related to the different lipid structures of the extracted lipids was suggested. To increase the knowledge of the specific role of different lipids on skin structure, a selective lipid modification is proposed. This study assess the effect of Ac on skin barrier lipids in-depth. Synchrotron based Fourier-transform infrared microspectroscopy (FTIR), which is used to study SC lipid organization, revealed a more ordered lipid organization after Ac treatment. *In vitro* experiments using Franz cells, which were selected to follow the SC barrier function capability, demonstrated that Ac-

treated skin retained caffeine and ibuprofen on the SC with very low permeation of both compounds into the deeper skin layers. *In vitro* transepidermal water loss (TEWL) measurements revealed the ability of Ac to induce a less water permeated skin. Although an important lipid fraction has been removed, Ac skin treatment brings to a skin where the remaining lipids promote an improved barrier function. These results could lead to a better understanding of the role of different lipid components in skin structure.

Abbreviations:

Ac	Acetone
AUC	Area under the curve
D	Dermis
E	Epidermis
FTIR	Fourier-transform infrared
HEX	Hexagonal
HPLC	High-performance liquid chromatography
LIQ	Liquid
LPP	Long periodicity phase
OTC	Optimal Cutting Temperature compound
OR	Orthorhombic
RF	Receptor fluid
RH	Room Humidity
SC	Stratum Corneum
SLS	Sodium laureth sulfate
SPP	Short periodicity phase
TEWL	Transepidermal water loss

Keywords: skin, lipids, TEWL, percutaneous absorption, permeability

1. Introduction

The skin is the outermost organ of the body, and it acts as a waterproof, insulating shield that guards the body against extreme temperatures, damaging sunlight and harmful chemicals. Skin also exudes antibacterial substances that prevent infection and manufactures vitamin D for converting calcium into healthy bones. Skin health depends on an intact skin barrier, which is provided by the outermost layer of the epidermis, the SC [1]. In the SC, anucleated keratin-filled corneocytes are dispersed in a lipid matrix, which consists of cholesterol, long-chain free fatty acids, and several classes of ceramides [1]. Lipids in the SC play an important role in skin barrier properties. As the continuous phase in the SC, intercellular lipids are the major transport route through the skin, and the barrier function highly depends on the SC lipid structure [2, 3]. The structural organization of the SC lipids has been widely studied [4, 5]; however, consensus about a unique model describing lipid packing does not exist [6]. Understanding of the lipid

matrix structure and molecular interactions dictating this structure is crucial to understand and control the penetration of substances through the skin.

Skin permeability strongly depends on the physical state of intercellular lipids. These lipids form the building blocks for proper lipid organization in the SC. The lipids form two lamellar phases with repeat distances of approximately 13 and 6 nm, which are referred to as the long periodicity phase (LPP) and short periodicity phase (SPP), respectively [7, 8]. The organization of lipids within the lamellae is referred to as lateral organization. There are three possible arrangements of the lipids: a very dense, ordered orthorhombic (OR) where aliphatic chains adopt all-*trans* conformation and are organized in a rectangular crystalline lattice; a less dense, ordered hexagonal (HEX) organization; or a disordered liquid (LIQ) organization where the chains exhibit a high degree of *gauche* isomerization and the lateral organization is entirely lost [9, 10].

The penetration of molecules through the SC depends on the order and packing of the lipid hydrocarbon chains. A highly ordered phase offers a higher resistance to molecule diffusion than a disordered phase [11]. The majority of the intercellular lipids are in a solid state at RH and ambient temperature. However, several studies indicate that a small fraction of skin lipids exist in a fluid state [12]. In fact, the most recent SC model proposed is based on a sandwich model consisting of two broad lipid layers with a crystalline structure separated by a narrow central lipid layer with fluid domains [13]. In a previous study the presence of compositionally different lipids domain when extracted with different solvents was demonstrated by differential scanning calorimetry [10]. Lipid unsaturations marked the differences between fluid and solids lipids. It is well recognized that the mobility (fluidity) of the different SC components is very important to several aspects of the vital functions of SC. The existence of fluid lipids is crucial to the barrier properties of the SC because these are lipids likely to constitute a major transport routes. Presumably, water and other small molecules that penetrate the SC diffuse through the fluid lipid regions as the permeability is considerably higher in the fluid phase than in the solid phase [14, 15]. The existence of lipids in ordered (OR and HEX) and disordered (LIQ) phases has been demonstrated in isolated SC [16]. An exhaustive analysis of band positions by ATR-FTIR has reported relevant data on the lipid organization of native SC, allowing for detection of alkyl chain vibrations of the lipids and consequently understanding of the lipid conformational order and alkyl chain packing [17, 18].

Transepidermal water loss (TEWL) is frequently used as an indicator of the integrity of skin barrier function. The measurement of TEWL allows total water loss to be determined from viable epidermis (E) and dermis (D), diffusing through the SC to the skin surface [19]. The relationship between TEWL and the degree of barrier disruption after skin damage is accepted as a measurement of barrier function efficacy. A less perfect SC with an increased TEWL is a more permeable barrier to drug penetration [20, 21]. Skin permeation can be studied using *in vitro* methods because it is possible to maintain the barrier properties of the SC in excised skin. *In vitro* percutaneous absorption studies are useful in identifying the depth that a given drug penetrates into the skin. *In vitro* studies using animals or human skin are an elegant tool to obtain data on the permeability of test substances through the skin and distribution across the different cutaneous compartments [22-25].

Organic solvents are frequently used as skin disruption agents, and topical treatment is attributed to the selective/integral depletion or the modification of intercellular skin lipids [26]. The mixture chloroform/methanol (2/1 v/v) depletes high levels of the SC lipids, giving rise to an important disruption of the skin barrier [27]. Some studies demonstrated that after Ac treatment, bulk lipids are removed from the SC. This effect was directly correlated with the degree of barrier perturbation [28-30]. Additionally, short-term Ac treatment acts differently because it does not extract significant levels of barrier lipids from normal human skin [31]. In a previous study, SC barrier function modification due to two different solvent systems was investigated. Different lipid structures of the solvent-extracted lipids accounted for different changes in the skin permeability [10].

The main aim of this work is to increase the understanding of the specific role of the different lipids on skin barrier function. Selective modification of skin lipids could induce a particular alteration of the

organizational state of the remaining lipids, leading to modifications of the physiochemical properties of the resultant tissues. Then, the effect of a selective modification of skin barrier lipids is being determined. To this aim, modification of Ac on skin barrier function was evaluated. The anionic surfactant SLS was also used as barrier function disruption procedure. FTIR microspectroscopy using synchrotron radiation enabled evaluation of SC lipid organization on the different skin layers. Moreover, *in vitro* permeation studies were performed to follow the SC barrier function capability.

2. Materials and Methods

2.1. Reagents

Acetone, and methanol (all for analysis) were purchased from Merck (Darmstadt, Germany). Caffeine (pure), bovine serum albumin, ibuprofen (pure), phosphate-buffered saline, gentamicin and sodium laureth sulfate (>98.5 %) were obtained from Sigma-Aldrich (St. Louis, Missouri, USA). Texapon N 70 was purchased from Basf (Ludwigshafen, Germany). Ethanol was obtained from Carlo Erba (Milan, Italy), and purified water was obtained from the Milli-Q plus 185 ultra-pure water system (Millipore, Bedford, MA, USA).

2.2. Skin samples

Porcine skin (three different donors) was obtained from the unboiled back of freshly sacrificed domestic pigs weighing 20–30 kg. The pig skin was provided by the Clinic Hospital of Barcelona, Spain. The management of the Landrace Large White pigs used in this study conforms to the Guide for the Care and Use of Laboratory Animals published by the United States National Institutes of Health [32]. The excised skin was dermatomed to 500 ± 50 μm thickness (Dermatome GA630, Aesculap, Tuttlingen, Germany). The discs of dermatomed skin were obtained using an iron punch (25 mm of diameter) in such a manner that they fit into the Franz diffusion cells. The skins discs were stored at -20 $^{\circ}\text{C}$ and thawed immediately for subsequent usage.

2.3. FTIR Experiments

Pig skin discs dermatomized to 500 μm ($\Phi=25$ mm) were used for infrared microspectroscopy experiments. Skin discs were soaked in 10 ml of Ac or 10 ml of the anionic surfactant SLS (1 %) for 4 h in a bath at 37 $^{\circ}\text{C}$. To avoid solvent evaporation, all treatments were conducted inside a closed vial. Control pig skin was submitted to the same treatment protocol with water. Finally, all skin samples were covered firstly with aluminium paper and then with Optimal Cutting Temperature compound (OCT, Bright Instruments, Bedfordshire, UK). The aluminum paper was used to avoid the direct contact between the skin and the OCT. Next, skin samples were frozen in liquid N₂ before cutting into transverse 6 μm thick sections. The cuts were performed using a Cryostat CM3050 (Leica Biosystems Nussloch, Germany). The different skin sections were placed on CaF₂ circular windows (13 mm in diameter) (Crystran Ltd., UK).

Infrared microspectroscopy was performed using synchrotron radiation at the MIRAS beamline at the ALBA synchrotron (Cerdanyola del Vallès, Spain). The infrared light source provided by the infrared beamline MIRAS achieves high lateral spatial resolution allowing subcellular molecular distribution analysis. The use of the narrow and intense infrared synchrotron light at infrared beamline MIRAS allows the biochemical investigation of different regions and structures of the skin. The FTIR measurements were performed using the 3000 Hyperion microscope coupled to Vertex 70 spectrometer. This microscope is equipped with a liquid nitrogen-cooled mercury cadmium telluride 50 microns MCT-A detector. The microscope used a 36X Schwarzschild objective (NA=0.65) and a matching 36X condenser.

Optical micrographs were produced from the microscope in the visible image mode to define the sampling positions. The spectra were acquired at room temperature in transmission mode. The data were recorded using a single masking aperture size of 10 μm x 10 μm . Spectra were collected in the 4000–900

cm⁻¹ mid-infrared range at a spectral resolution of 4 cm⁻¹ with 128 co-added scans per spectrum. A minimum of 50 spectra were collected for each studied skin sample.

Data treatment

FTIR data treatment and statistical analysis were done with two different programs OPUS (Bruker, USA) and Unscrambler® (CAMO software, Norway) [33]. To extract spectral information, the obtained data was processed as follows: Each spectrum was manually grouped considering in which layer was acquired (stratum corneum, epidermis or dermis) and if it was treated (Ac or SLS) or not (water). Spectra were baseline corrected in the region of interest and then normalized by unit vector. The second derivative was calculated using the Savitzky-Golay algorithm using third polynomial order and 7 smoothening points. The normalized reduced average spectrum and its corresponding Savitsky-Golay second derivative (third polynomial order and 7 smoothening points) for each data set, in each corresponding spectral defined group, were obtained. To compare the different spectra groups, the principal component analysis were performed. Whenever the PCA captured a variation in the data sets in the defined spectral ranges, the corresponding loadings plot, the average and/or second derivative spectra were carefully examined to find the subtle spectral differences responsible for this variation

2.4. Transepidermal water loss (TEWL) measurements

A study was performed to simulate the *in vitro* barrier function modification measured by evaluations of the skin TEWL with a Tewameter TM210 (Courage & Khazaka, Cologne, Germany). Several skin discs were positioned in a simple model Franz static diffusion cells (3 ml, 1.86 cm², Lara-Spiral, Courtenon, France). The Franz cells consisted of an upper donor compartment and receptor chamber (3 ml of volume). The skin discs were mounted in the Franz cell with the SC side on top. The receptor chamber was filled with a receptor fluid containing 1 % bovine serum albumin and 0.04 % gentamicin sulfate in phosphate-buffered saline (pH 7.4) aqueous solution. The cells were placed in a thermostated water bath (Julabo, Seelbach, Germany) assuring a skin surface temperature of 32°C while being subjected to magnetic stirring (Variomag 15 and Telemodul, H+P Laortechnik GmbH, Germany). The bath has a platform located on the bath water where the Franz cells are located. This platform acts as a barrier and assures that there is not water flux coming from the water bath that could interfere with the TEWL measurements. After stabilization of the cells for 1 h, the measurement of TEWL was performed. Skin discs with TEWL values less than 15 g m⁻² h⁻¹ were considered acceptable [34]. In this *in vitro* assay, a more realistic treatment was chosen. Briefly, 90 µl of Ac or SLS (1 %) was applied to the skin surface and left occluded to avoid solvents evaporation for 2 h. Water (90 µl) was used as a control. Six skin samples were required for each treatment. At the end of the exposure period, the skin samples were left uncovered for 2 h to assure skin surface temperature stabilization (32 °C) and the evaporation of the treatment previously performed. Then, the skin integrity was evaluated by measuring the TEWL [35].

2.5. *In vitro* percutaneous assay

The OECD guidelines and the published opinions of the Scientific Committee on Consumer Safety were closely adhered to during this study [22-25]. The permeation behaviour of each compound was examined *in vitro* using glass Franz diffusion cells, also used for TEWL measurements. In this percutaneous assay the receptor solution was specific for each studied compound and based on its own solubility. For caffeine, the receptor solution was made of 1 % bovine serum albumin and 0.04 % gentamicin sulfate in phosphate-buffered saline (pH 7.4) aqueous solution. Ibuprofen absorption study was performed with phosphate-buffered saline (pH 7.4) aqueous solution as the receptor fluid. After maintaining the Franz cells with the skin discs at 37 °C, to ensure that the skin surface was at 32 ± 1 °C, the integrity of each skin sample was assessed by determining the TEWL. Skin discs with TEWL values less than 15 g m⁻² h⁻¹ were considered acceptable. Similar to the TEWL measurements section, 90 µl of Ac (100 %), SLS (1 %) or water was applied to the entire surface before the diffusions experiments, and the skin samples were left occluded for 2 h. Four skin samples were required for each treatment. At the end of the exposure period, the skin samples were left uncovered for 1 h to assure skin surface temperature stabilization (32 °C) and the evaporation of the treatment previously performed. Then, TEWL values were determined again to verify a correct skin sample condition. Next, 10 µl of ethanol-water (1:1) solution of caffeine (1

%) or ibuprofen (1%) were applied to the entire skin surfaces. After an exposure time of 24 h, the test formulation remaining in the skin were removed by the following specific wash: first with 0.5 ml of Texapon solution (0.5 %, v/v) and then twice with distilled water (2 x 0.5 ml). Subsequently, the skin was dried with a cotton swab. Water aliquots from all tips and cotton swabs used as well of the top of the cell were pooled, which constituted the fraction of active compound that remained on the surface. Then, the receptor fluid (RF) was removed from the receptor compartment and brought to 5 ml in a volumetric flask. The surface horny layers of the SC were removed from the treated area of each skin sample by eight successive tape strippings wherein adhesive tape (D-Squame, Cuderm Corp., Dallas, TX, USA) was applied under controlled pressure (80 g/cm² for several seconds). The tape strips were grouped as strips 1-4 and strips 5-8 and placed in glass vials. Then, the E was separated from the D after heating the skin at 80 °C for a few seconds. The D was subsequently cut into small pieces.

The amount of caffeine or ibuprofen in the different layers and washing solution was extracted with methanol-water solution (50:50) or methanol 100% respectively overnight. Afterward, all samples were shaken for 30 minutes at room temperature. The D, E and SC samples were sonicated for 15 minutes. Before the analytical determination of the samples, they were filtered through a 0.45-µm Nylon Cameo filter (Scharlab, Spain) and analysed using high-performance liquid chromatography (HPLC).

2.5.1. High-performance liquid chromatography analysis

Caffeine and ibuprofen were determined in all samples by HPLC using Hitachi-Merck HPLC equipment (Darmstadt, Germany). The apparatus is equipped with a variable UV-Vis detector (L-4250) and C18 column (LiChrocart 250-4/LiChrosorb RP-18.5 µm, Darmstadt, Germany). The mobile phase used for caffeine was a mixture of methanol and water (20:80 v/v) at a 1 ml/min flow rate. The caffeine was measured at 271 nm and the retention time was 4.5 min. In the case of ibuprofen the mobile phase was MeOH:H₃PO₄ at 1.2 % (67:33, v/v) with a flow of 1 ml/min. The wavelength used was 221 nm and the retention time obtained was 17 min. The area under the peak was used to calculate the concentration of both compounds with external standards that exhibited linearity over the concentration range indicated in Table 1. The intraday and interday variations of both methods were less than 2 %. Table 1 presents the analytical results obtained in both validations.

	Caffeine	Ibuprofen
Detection limit (µg/ml)	0.079 ± 0.030	0.448 ± 0.021
Quantification limit (µg/ml)	0.222 ± 0.015	0.896 ± 0.107
Calibration curve	Area _{271nm} = 19156[caffeine conc.]- 2414.8	Area _{264nm} = 319598[ibuprofen conc.]+46268
Linearity coefficient (r ²)	0.9999	0.9998
Repeatability (CV _{intra})	0.47 ± 0.11	3.64 ± 1.05
Intermediate precision (CV _{inter})	1.20 ± 0.23	5.03 ± 1.50

Table 1. Data obtained for the analytical validation of caffeine and ibuprofen

2.6. Statistical Analysis

Statistical analyses were performed with StatGraphics software (version 5.0). Differences were tested for statistical significance using non-parametric tests (Kruskal-Wallis) with p<0.05 considered as significant.

3. Results

3.1. FTIR Experiments

FTIR spectra for untreated and treated skin samples in the different skin layers were recorded in the range of 4000–900 cm⁻¹. Synchrotron-based FTIR measurements at the infrared beamline MIRAS allowed obtaining FTIR chemical-maps of the skins sections. The absorption bands from 2800 to 3000 cm⁻¹ correspond to the CH₂ stretching frequencies, and can be used to qualitatively map the distribution of the lipids in the different skin layers (Figure 1). The colour scale, from blue to pink, represents the location

and amount of the lipids. Pink-red areas concentrate the highest amount of the lipids while in blue areas no presence of these molecules is detected. Yellow and green represent areas with intermediate amounts.

For each skin section evaluated, the three skin layers can be distinguished in the map: stratum corneum (SC), epidermis (E) and dermis (D) (Figure 1). The image on normal skin, treated with water, shows the main intensity lipids signals on the SC layer as it could be expected. Both skin treated, SLS and Ac, shows a more irregular lipid distribution among the different skin layers.

Figure 1

The averaged spectra of the different skin layers in the maps above were extracted and analysed individually. Figure 2a presents the average absorbances for the asymmetric and the symmetric stretching CH_2 vibrations that occurred at 2920 and 2850 cm^{-1} near the skin surface (including SC and E layers) for the three skin samples evaluated. In all the spectra obtained, both stretching bands were intense and did not overlap with other vibrational modes. The second derivative of the spectra were used for determining peak positions and were statistically analyzed by means of principal component analysis (PCA) (Figure 2 and Table 2).

	CH_2 asym stretching (cm^{-1})	CH_2 sym stretching (cm^{-1})	CH_2 asym/ Amide II ratio	CH_2 sym/ Amide II ratio
Skin Water	2921.63	2850.20	2.19	0.96
Skin SLS	2923.55	2850.20	2.07	0.93
Skin Ac	2919.69	2848.72	1.65	0.83

Sodium laureth sulfate (SLS), acetone (Ac)

Table 2. The symmetric (sym) and asymmetric (asym) CH_2 stretching peaks frequencies (cm^{-1}) obtained from second derivatives spectra and the mean values of the $\text{AUC}_{\text{CH}_2 \text{ asym or sym}} / \text{AUC}_{\text{amide II}}$ ratio for the skin surface treated with water and lipid-modified skin, with acetone (Ac) and the surfactant (SLS).

FTIR results show differences when evaluating the CH_2 stretching of the skin that underwent surfactant treatment compared with the water-treated sample. Table 2 shows FTIR absorption peaks values for the symmetric and asymmetric stretching modes. Concerning the asymmetric mode, in the SLS treated sample, the band shifted to higher wavenumbers values indicating thus a more disordered lipid organization [17]. However, when the porcine skin was treated with Ac, a decrease of the CH_2 asymmetric frequency was observed, reaching smaller values than that detected in the water-treated skin. These results indicate a major ordering of the skin lipids in the sample treated with Ac compared to the control sample. The stretching symmetric vibrations of CH_2 are more sensitive than the asymmetric vibrations to packing changes [36]. The results are also presented in Table 2. Very similar symmetric CH_2 bands were observed for water-treated and SLS-treated skin samples. Nevertheless, the skin treated with Ac exhibited a decrease of the vibrational band position for the symmetric CH_2 from 2850.20 to 2848.72 cm^{-1} , indicating a transition from HEX to OR, a more ordered phase. PCA scores of the points along the PC-1 and PC-2 axis confirms the significant different behaviour of Ac treated skin lipids when compared with both SLS and water treated skins (Figure 2c).

Figure 2

Moreover, semi-quantitative analysis of the skin lipids can also be performed calculating the mean ratio for $\text{AUC}_{\text{CH}_2 \text{ asym or sym}} / \text{AUC}_{\text{amide II}}$ and these results are shown in Table 2. To overcome problems that could be associated with different thickness of the skin, and assuming that amides are homogeneously distributed,

amide II vibration was used to normalize the AUC obtained from the CH₂ symmetric and asymmetric stretching modes. This quantification was performed in all the maps obtained by FTIR for all the different treated skin samples. The relative concentration of lipids was higher for the skin treated only with water being the acetone treated skin the one with a lower concentration of lipids present in its structure.

In addition to the CH₂ stretching bands, the CH₂ scissoring region, studied between 1480 and 1460 cm⁻¹, are known to provide information about the lateral packing of the skin lipids [17]. The scissoring region is rather small, therefore, this region was enlarged (1417-1480 cm⁻¹) and the second derivative of this spectral region was used to facilitate analysis of the most significant peaks (Figures 3a and 3b). The PCA score gives a pattern for the three samples which are separated in three different groups, indicating differences in the chemical structure of this spectral region. The second derivative shows that the skin treated with Ac presents an increased band at 1471 cm⁻¹ which can be compatible with an OR packing, corresponding to one of the bands of the characteristic OR symmetric split [36, 37]. The occurrence of OR phases is revealed by the splitting of the scissoring modes to produce a doublet with components at ~1473, 1463 cm⁻¹.

Then, it is important to remark that even the lipid solubilization detected for the Ac treated skin sample, the remaining lipids presents a very ordered structure corresponding to OR packing.

Figure 3

3.2. Transepidermal water loss (TEWL) measurements

Evaluations of TEWL on differently modified skin samples were followed by a study of the *in vitro* barrier function alteration. The results are presented in Table 3. When the skin undergoes an occluded treatment with water, the barrier function is somewhat modified with an increased TEWL value. When the skin is modified with SLS, an increase in TEWL is also found, indicating an alteration in barrier function due to the anionic surfactant. Surprisingly, the modification of the skin with Ac did not increase the TEWL value as found with the other skin treatments. In contrast, a statistically significant decrease in this parameter was found. These results are consistent with the previous section and suggested an improvement in barrier function in response to Ac treatment.

	Skin Water	Skin SLS	Skin Ac
Initial TEWL (g/m²/ hour)	9.69 ± 1.77	9.34 ± 2.59	10.80 ± 1.69
Final TEWL (g/m²/ hour)	11.42 ± 2.41	19.85 ± 11.57	8.46 ± 3.54
%	20.03 ± 27.78	112.40 ± 94.51	-23.75 ± 20.53*

Sodium laureth sulfate (SLS), acetone (Ac)

Table 3. TEWL values (mean values ± SD with n=6) before (initial) and after (final) the different skin treatments (water treated (water) and lipid-modified SC with the surfactant (SLS) and acetone (Ac)) expressed as g/m²/ hour. TEWL variation (%) were evaluated versus the initial TEWL values. (*p<0.05, statistics between the skin treated with water and the skin treated with the different disrupting agents).

3.3. *In vitro* percutaneous assay

The in vitro percutaneous absorptions of caffeine and ibuprofen were investigated on skin discs subject to different barrier disturbance treatments. These compounds were chosen due to their different hydrophilic properties being at skin physiological pH (pH 5) caffeine clearly more hydrophilic than ibuprofen. The percutaneous absorption assay used in this research determined the distribution of both compounds in the various skin compartments and the receptor fluid mimicking the systemic compartment. The mean caffeine or ibuprofen content expressed in $\mu\text{g}/\text{cm}^2$ in the different skin compartments, skin surface (excess), and RF are presented in Tables 4 and 5. Furthermore, the global percutaneous absorption values and the total recovery found during the experiments are presented. The resulting mass balance was acceptable for each evaluated skin after an exposure time of 24 h being always superior to 80%. The concentrations found in the epidermis (except for the SC) and dermis can be absorbed and can reach the systemic level. The amount of percutaneous absorption obtained for each compound is the sum of the concentrations in the epidermis, dermis, and receptor fluid (E + D + RF).

	Skin Water	Skin SLS	Skin Ac
W	46.76 $\pm$ 1.97	17.18 $\pm$ 8.63 **	46.14 $\pm$ 2.66
SC	1.35 $\pm$ 0.61	0.69 $\pm$ 0.26**	1.25 $\pm$ 0.21
E	0.42 $\pm$ 0.27	0.74 $\pm$ 0.12**	0.46 $\pm$ 0.08
D	0.25 $\pm$ 0.14	0.82 $\pm$ 0.16 **	0.15 $\pm$ 0.03
RF	3.22 $\pm$ 0.74	25.45 $\pm$ 7.55**	1.65 $\pm$ 0.61 **
Total Recovery (%)	51.99 $\pm$ 1.32 (97.38 $\pm$ 2.48)	44.87 $\pm$ 1.52 ** (84.05 $\pm$ 2.84)	49.66 $\pm$ 3.23 (93.00 $\pm$ 6.04)
Percutaneous absorption	3.89 $\pm$ 0.95	27.00 $\pm$ 7.64 **	2.26 $\pm$ 0.72 *

Sodium laureth sulfate (SLS), acetone (Ac), surface excess (W), stratum corneum (SC), epidermis (E), dermis (D) and receptor fluid (RF)

Table 4. Distribution of caffeine (mean values $\pm$ SD with n=4) in the different skin compartments, skin surface (excess), and RF as well as percutaneous absorption values (E + D + RF), expressed as $\mu\text{g}/\text{cm}^2$. (**p<0.05 and *p<0.10 statistics between the water treated skin and the skin treated with the different disruption agents).

The penetration profile of both compounds applied to the control skin, submitted only to water occlusion, indicates a much lower penetration of caffeine into each skin layer and in the receptor fluid, giving up to a percutaneous of about 4 $\mu\text{g}/\text{cm}^2$ compared to 9 $\mu\text{g}/\text{cm}^2$ of ibuprofen. These results could be expected due to the known favoured skin permeability of hydrophobic compounds. The influence of the treatments can also be envisaged. The percutaneous absorption of both studied compounds was significantly increased for the skin that was previously treated with the SLS surfactant, caffeine (27 $\mu\text{g}/\text{cm}^2$) and ibuprofen (37 $\mu\text{g}/\text{cm}^2$), compared with water-treated skin (4 and 9 $\mu\text{g}/\text{cm}^2$ respectively). Again, the results from the skin previously treated with Ac demonstrated the different behaviour of this solvent when applied to the skin surface. A lower amount of permeated ibuprofen (14 $\mu\text{g}/\text{cm}^2$) when compared with the other skin disturbance treatment was obtained. Besides, the significant diminution of skin percutaneous absorbed caffeine (2 $\mu\text{g}/\text{cm}^2$) when compared to both control and SLS treated skin suggests an adequate skin barrier function after Ac skin treatment. This indicates that acetone treatment promotes a skin permeation of a lipophilic compound similar to the control skin and prevents permeation of a hydrophilic compound even in a greater extent to the control skin.

	Skin Water	Skin SLS	Skin Ac
W	39.42 $\pm$ 15.50	14.70 $\pm$ 9.35 **	34.15 $\pm$ 1.55
SC	10.15 $\pm$ 1.47	2.45 $\pm$ 1.06 **	9.09 $\pm$ 1.92
E	0.79 $\pm$ 0.33	0.44 $\pm$ 0.38**	1.68 $\pm$ 0.36
D	0.68 $\pm$ 0.44	2.21 $\pm$ 0.91**	0.81 $\pm$ 0.25
RF	7.56 $\pm$ 0.84	35.40 $\pm$ 10.78**	11.19 $\pm$ 2.72
Total Recovery (%)	59.21 $\pm$ 14.36 (91.21 $\pm$ 27.70)	55.22 $\pm$ 6.24** (81.83 $\pm$ 12.01)	56.92 $\pm$ 2.58 (83.79 $\pm$ 2.63)
Percutaneous absorption	9.03 $\pm$ 1.18	37.05 $\pm$ 10.23**	13.68 $\pm$ 2.78

Sodium laureth sulfate (SLS), acetone (Ac), surface excess (W), stratum corneum (SC), epidermis (E), dermis (D) and receptor fluid (RF)

Table 5. Distribution of ibuprofen (mean values $\pm$ SD with $n=3$) in the different skin compartments, skin surface (excess), and RF as well as percutaneous absorption values ($E + D + RF$), expressed as $\mu\text{g}/\text{cm}^2$. (** $p < 0.05$ statistics between the water treated skin and the skin treated with the different disruption agents).

After 24 h of exposure, both studied compounds were mainly located in the SC layer on the skins that were previously treated with water and Ac. Both skins exhibited a strong barrier function that could retain in the skin surface (W sample) more than 50 % of all applied ibuprofen and approximately 90 % of all deposited caffeine. Moreover, the concentration of caffeine and ibuprofen in E and D significantly decreased, demonstrating the presence of slight skin penetration of both compounds. A different behaviour was noted for skins that were pre-treated with SLS. In these skins, a significantly reduced level of caffeine and ibuprofen was observed in the SC layer and it is increased in the dermis which indicates high penetration of both compounds and can be attributed to disrupted SC barrier function. Furthermore, results of caffeine and ibuprofen concentrations in the fluid receptor indicate that SLS-treated samples induce a significantly high level of permeated compounds. On the contrary, it is important to remark the lowest amount of caffeine in the receptor fluid in the case of acetone treated skin even lower than the one obtained for the control skin.

4. Discussion

The skin barrier is located in the uppermost layer of the skin, namely, the SC, and functions to protect the body from excessive TEWL and prevent the penetration of compounds into the body via the epidermis. The SC consists of corneocytes with lipids in the intercellular regions. Commonly used laboratory procedures to remove intercellular SC lipids include solvent systems, such as chloroform/methanol. Typically, these solvents systems extract all intercellular SC lipids. For a more specific study on the influence of the different lipids on the barrier properties of the SC, more selective lipid depletion is proposed. In a previous study, partial lipid depletion via Ac treatment on skin was demonstrated. A preferential solubilization of fluid lipids was envisaged, which could explain the decrease in the SC water permeability when submitted to Ac extraction [10]. This study sought to perform an in-depth assessment of the possibility of monitoring the SC lipid organization. Images of control skin, treated only with water, and skin treated with Ac and SLS showed different distribution of the skin lipids. The largest changes on the amount of lipids of the skin samples were attributed to the Ac treatment. FTIR microspectroscopy using synchrotron radiation is also used in this work to evaluate the skin barrier lipid organization. The SC intercellular lipid organization is associated with skin barrier function conditions. There are three possible arrangements of the lipids: a very dense, ordered OR organization; a less dense, ordered HEX organization; or a disordered LIQ organization [9, 10, 18, 31]. Despite the decrease on the amount of skin lipids, after Ac treatment, a clear diminution of the vibrational frequency of the FTIR stretching bands was noted. These results clearly indicate that after lipid extraction with Ac, the remaining intercellular SC lipids adopt a significant more organized structure corresponding to an OR phase, and this structure could be related to a lower skin permeability. FTIR measurements for the scissoring vibrations confirm that when the skin is treated with Ac the remaining lipids present a more ordered lateral packing organization. FTIR results agreed with previous SC lipid organization investigations using Confocal-Raman spectroscopy [10] where the ratio of peak heights I_{2880}/I_{2850} , which depend on the lateral order of the extended chains, was used as a qualitative measure of the lateral order. When skin previously treated with the Ac was compared with non-treated skin, a significant increase in the I_{2880}/I_{2850} ratio was observed, suggesting that the remaining lipids might adopt a more ordered conformation.

TEWL is predictive of the skin permeability barrier [21, 38]. A clear increase in this parameter was observed after SLS disruption process, demonstrating the damaging effect of the surfactant treatment on the skin barrier function. Instead, Ac treatment decreased the TEWL values obtained in all the experiments conducted, suggesting a modification of the skin permeability with a possible improvement in the skin barrier function. TEWL results agreed with previous permeability investigations using sorption experiments [10] where acetone treatment lead to a SC sample with a decreased diffusion coefficient which implied a decreased SC permeability.

Some researchers established the existence of a correlation of TEWL and percutaneous absorption [39]. Removing SC lipids should increase the percutaneous absorption of drugs. A percutaneous absorption study was developed on different treated skin samples with two topical actives with different hydrophilic properties. Much of the difficulty in predicting the absorption of a compound is due to the multilayered structure of skin, with regions varying in their capacity to solubilize hydrophilic and lipophilic substances. Skin absorption of a drug may be predicted by its physicochemical properties, including molecular weight and lipophilicity, which play a major role in the skin permeation process [40]. In this study, the absorption of both hydrophilic and a lipophilic compounds on modified skin is investigated. Drug distributions in the different skin layers could account for differences in skin lipid organization. Caffeine was selected as a standard chemical model used for skin absorption assays [41] and in cosmetics [42, 43]. Furthermore, ibuprofen was also chosen to evaluate percutaneous assay of a hydrophobic drug as ibuprofen permeability through human skin has been widely studied [44, 45].

Profile of studied compounds that permeated through the skin of our control sample, the non treated, or water treated, are consistent with results of previous investigations [35, 46]. The amount of caffeine and ibuprofen detected on the skin surface (W sample) was significantly higher for the skins previously treated with water and Ac. This is an expected result because although water can somewhat modify the skin structure, the derived damaged should not promote increased drug skin permeability. However, Ac is a solvent that can extract part of skin lipids [29, 30], thus, the ability of Ac-treated skin to retain a significant amount of caffeine or ibuprofen on the skin surface suggests that the remaining skin lipids could still act as a viable barrier function. The disorganization of the intercellular skin lipids on SLS treatment was confirmed with a significant decrease in the amount of both tested compounds found on the skin surface.

SLS treatment promotes skin lipid disorganization, leading to significant permeation of caffeine and ibuprofen into the deeper skin layers. The interaction of Ac with the skin lipids gave rise to a caffeine and ibuprofen penetration profiles similar to that observed for control skin. In both cases, effective skin barrier function demonstrated by a significant increase in both tested compounds on SC revealed reduced penetration into the deeper skin layers. These results clearly show that pre-treatment of skin with Ac gives rise to skin in which the remaining lipids can adopt a solid structure, improving barrier function.

Much penetration was obtained for the most hydrophobic compound ibuprofen related to the hydrophilic one caffeine in the control skin. It is important to remark that those differences were maintained for treated skin. Moreover, the lowest amount of caffeine found in the receptor fluid of the acetone treated skin indicates that acetone modification strongly prevents penetration of hydrophilic than hydrophobic compounds.

5. Conclusions

Earlier investigations envisioned that the differences on the skin lipid-extracted structures could account for different changes in skin permeability. In this study, FTIR measurements revealed that even an important lipid extraction was obtained after Ac treatment a more ordered SC lipid organization, which might reinforce the skin microstructure, was observed. This finding was confirmed in the *in vitro* permeability assays. The TEWL measurements demonstrate the ability of Ac to give rise to skin with decreased TEWL, thus improving barrier function. Additionally, the percutaneous absorption profile of a hydrophobic and a hydrophilic drug clearly demonstrated that Ac-treated skin retained both compounds on the SC, and minimal permeation into the deeper skin layers was noted. The assumed preferential interaction of Ac versus fluid lipids results in skin in which, although an important lipid fraction has been

removed, the remaining lipids promote an improved barrier function. The obtained results could lead to a better understanding of the role of the different lipid components on skin structure.

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Figures

Figure 1. FTIR-maps of the lipids vibrations modes distributions obtained from the skin treated with water (a) and lipid-modified skin, with acetone (b) and the surfactant (SLS) (c).

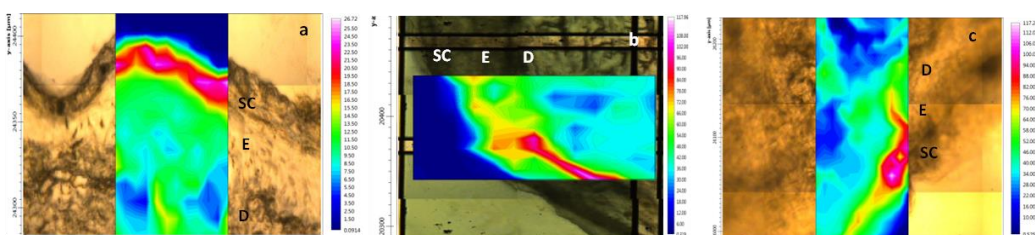


Figure 2 (a) Averaged FTIR spectra in the stretching region range 3000-2800 cm^{-1} of the epidermis ((A) asymmetric (asym) and (B) symmetric (sym) CH_2 stretching peaks), unit vector normalized (b) range normalized average second derivative spectra of lipids stretching region (c) Single points for PCA analysis. Skin treated with water (black) and lipid-modified skin, with acetone (red) and the surfactant SLS (green).

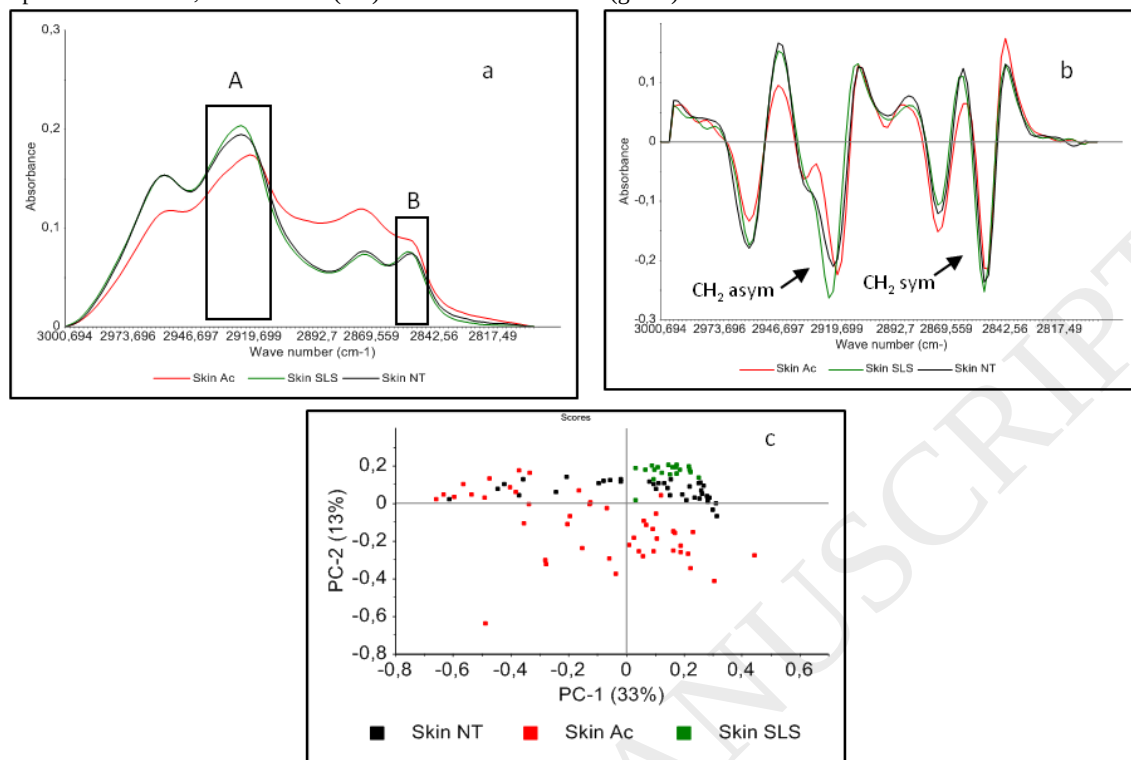


Figure 3 (a) averaged FTIR spectra in the scissoring region range 1417-1480 cm^{-1} of the epidermis, unit vector normalized (b) range normalized average second derivative spectra of lipids scissoring region (c) Single points for PCA analysis. Skin treated with water (black) and lipid-modified skin, with acetone (red) and the surfactant SLS (green).

